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United States Patent	12406341
Kind Code	B2
Date of Patent	September 02, 2025
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Correcting images for an ophthalmic imaging system

Abstract

In certain embodiments, an ophthalmic system images an eye region comprising at least one eye. The system includes a camera system and a computer. The camera system includes cameras that yield image portions of the eye region. Each camera is located at a position relative to the eye region and yields an image portion. The computer receives the image portions from the camera system. A first image portion is provided by a first camera, and a second image portion is provided by a second camera. The computer identifies target pixels of the first image portion, where the target pixels image a location of the eye region; determines image information of correction pixels of the second image portion, where the correction pixels image the same location of the eye region; and corrects the target pixels using the identified image information.

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Appl. No.:	18/615090
Filed:	March 25, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240331113 A1	Oct. 03, 2024

Related U.S. Application Data

us-provisional-application US 63492656 20230328

Publication Classification

Int. Cl.: G06T5/77 (20240101); A61B3/113 (20060101); A61B3/14 (20060101); G06T7/00 (20170101); G06T7/292 (20170101)

U.S. Cl.:

CPC G06T5/77 (20240101); A61B3/113 (20130101); A61B3/14 (20130101); G06T7/0002 (20130101); G06T7/292 (20170101); G06T2207/10012 (20130101); G06T2207/30041 (20130101); G06T2207/30196 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9172871	12/2014	Zhao	N/A	G06T 7/80
11487097	12/2021	Park	N/A	G02B 21/16
11792538	12/2022	Venkataraman	348/222.1	H04N 5/265
11984068	12/2023	Lee	N/A	G09G 3/3208
2014/0232830	12/2013	Ichige	348/47	H04N 13/239
2014/0321771	12/2013	Reinisch	382/284	G06T 11/60
2016/0263402	12/2015	Zhang	N/A	A61N 5/1049
2017/0234976	12/2016	Grauer	356/5.04	G01S 17/18
2018/0096487	12/2017	Nash	N/A	H04N 23/45
2020/0139889	12/2019	Christmann	N/A	B60R 1/002
2020/0150052	12/2019	Lee	N/A	G01N 21/8903
2020/0201022	12/2019	Shameli	N/A	G02B 23/243
2021/0243386	12/2020	Park	N/A	H04N 5/2628
2021/0272302	12/2020	Yang	N/A	G01B 11/22
2023/0230387	12/2022	Herman	701/25	G06V 20/56
2023/0342894	12/2022	Scharfenberger	N/A	G06T 3/4038
2024/0175244	12/2023	Bhupatiraju	N/A	G06T 7/12
2024/0257691	12/2023	Jeon	N/A	G09G 3/006
2024/0311964	12/2023	Li	N/A	G06T 3/4053
2024/0331113	12/2023	Abraham	N/A	G06T 5/77
2024/0331172	12/2023	Abraham	N/A	A61B 3/14
2024/0412506	12/2023	Kimita	N/A	G06V 10/993
2025/0017461	12/2024	Abraham	N/A	A61B 3/145
2025/0037418	12/2024	Momiyama	N/A	H04N 23/11
2025/0134374	12/2024	Vaziri	N/A	A61B 5/1128

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2016201457	12/2016	AU	A61B 3/117
2016201457	12/2016	AU	A61B 3/117

2851687	12/2012	CA	A61B 3/0025
3236064	12/2022	CA	A61B 3/12
104768447	12/2014	CN	A61B 3/0008
113286078	12/2020	CN	H04N 5/23222
113592777	12/2020	CN	G06T 5/50
116148262	12/2022	CN	G01N 21/8851
4131182	12/2022	EP	A61B 3/0008
2023027405	12/2022	JP	A61B 34/10

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to ophthalmic imaging systems, and more particularly to correcting images for ophthalmic imaging systems.

BACKGROUND

(2) Ophthalmic systems often provide an image of an eye for diagnosing or treating the eye. The eye is typically illuminated to generate the image. However, the illumination may cause undesired reflections that appear in the image.

BRIEF SUMMARY

(3) In certain embodiments, an ophthalmic system images an eye region comprising at least one eye. The system includes a camera system, an illuminator, and a computer. The camera system includes cameras that yield image portions of the eye region. Each camera is located at a position relative to the eye region and yields an image portion. The illuminator is located at a position relative to the eye region and directs light towards the eye region. The computer receives the image portions from the camera system. A first image portion is provided by a first camera, and a second image portion is provided by a second camera. The computer identifies reflection pixels of the first image portion, where the reflection pixels image a reflection of light from the illuminator reflected by a location of the eye region; determines image information of correction pixels of the second image portion, where the correction pixels image the same location of the eye region; and corrects the reflection pixels using the identified image information to reduce the reflection.

(4) Embodiments may include none, one, some, or all of the following features: The cameras comprise stereoscopic cameras arranged symmetrically about the system axis of the camera system. The computer identifies reflection pixels of the first image portion by detecting light saturated pixels as the reflection pixels. The computer identifies reflection pixels of the first image portion by identifying the reflection pixels according to the position of the illuminator relative to the eye region and the position of the first camera relative to the eye region. The computer identifies reflection pixels of the first image portion by: receiving eye tracking information describing movement of the eye; and identifying the reflection pixels according to the previous location of the reflection pixels and the movement of the eye. The first camera is distinct from the second camera. The first camera is at a first position relative to the eye region, and the second camera is at a second position relative to the eye region distinct from the first position. The first camera is the same as the second camera, and the second image portion provided prior to the first image portion. The second image portion comprises default eye image information. The computer corrects the reflection pixels using the identified image information by replacing the reflection pixels with the correction pixels. The computer corrects the reflection pixels using the identified image information by applying an averaging function to the reflection pixels and the correction pixels. The computer corrects the

reflection pixels using the identified image information by: generating a correction overlay using the correction pixels; and placing the correction overlay over the reflection pixels.

(5) In certain embodiments, an ophthalmic system images an eye region comprising at least one eye. The system includes a camera system and a computer. The camera system includes cameras that yield image portions of the eye region. Each camera is located at a position relative to the eye region and yields an image portion. The computer receives the image portions from the camera system. A first image portion is provided by a first camera, and a second image portion is provided by a second camera. The computer identifies target pixels of the first image portion, where the target pixels image a location of the eye region; determines image information of correction pixels of the second image portion, where the correction pixels image the same location of the eye region; and corrects the target pixels using the identified image information.

(6) Embodiments may include none, one, some, or all of the following features: The cameras comprise stereoscopic cameras arranged symmetrically about the system axis of the camera system. The computer identifies target pixels of the first image portion by detecting light saturated pixels as the reflection pixels. The computer identifies target pixels of the first image portion by: receiving eye tracking information describing movement of the eye; and identifying the target pixels according to the previous location of the target pixels and the movement of the eye. The first camera is distinct from the second camera. The first camera is at a first position relative to the eye region, and the second camera is at a second position relative to the eye region distinct from the first position. The first camera is the same as the second camera, and the second image portion provided prior to the first image portion. The second image portion comprises default eye image information. The computer corrects the target pixels using the identified image information by replacing the target pixels with the correction pixels. The computer corrects the target pixels using the identified image information by: applying an averaging function to the target pixels and the correction pixels. The computer corrects the target pixels using the identified image information by: generating a correction overlay using the correction pixels; and placing the correction overlay over the target pixels.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates an example of an ophthalmic system with an eye camera that provides images of an eye region, according to certain embodiments

(2) FIG. 2 illustrates an example of the field of view (FOV) of the camera system of FIG. 1, according to certain embodiments;

(3) FIGS. 3A and 3B illustrate examples the camera system of FIG. 1 tracking eye regions, according to certain embodiments; and

(4) FIGS. 4 and 5 illustrate an example of a method for correcting pixels in an eye image, according to certain embodiments. FIG. 4 shows image portions that are used to yield a corrected image, and FIG. 5 shows a flowchart of the method.

DESCRIPTION OF EXAMPLE EMBODIMENTS

(5) Referring now to the description and drawings, example embodiments of the disclosed apparatuses, systems, and methods are shown in detail. The description and drawings are not intended to be exhaustive or otherwise limit the claims to the specific embodiments shown in the drawings and disclosed in the description. Although the drawings represent possible embodiments, the drawings are not necessarily to scale and certain features may be simplified, exaggerated, removed, or partially sectioned to better illustrate the embodiments.

(6) According to a known technique for reducing problematic pixels of an image such as reflections, the illumination is polarized, and the polarized component is filtered out from the

image. However, diffuse surfaces reflect polarized light in different directions, which can cause problems, e.g., the surface of an open LASIK flap can yield undesirable color shimmers. According to another known technique, reflections can be removed by interpolating information from adjacent pixels. However, known interpolation techniques may yield a less accurate image.

(7) The ophthalmic imaging systems described herein have multiple cameras that provide images of the eye from different viewing directions to generate a digital image, such as an image in a microscope. The cameras have different lines of sight, so problematic pixels, such as reflections, appear at different locations in the images of the eye. If there is a reflection at a location of the eye in one image, image data from another image where the reflection is absent at that location may be used to correct the image. If there are locations where no image provides non-reflection image data, data from surrounding pixels may be interpolated to provide information for these locations.

(8) FIG. 1 illustrates an example of an ophthalmic system **10** with an eye camera **12** that provides images of an eye region **14** (which may include one or both eyes of a patient), according to certain embodiments. For ease of explanation, certain eye features are used to define an example coordinate system **16** (x, y, z) of the eye. For example, the eye has a center (e.g., pupil center, apex, vertex) and an eye axis **15** (e.g., optical or pupillary axis) that can define the z-axis of eye coordinate system **16**, which in turn defines an xy-plane of system **16**. Eye region **14** has a region axis **17**. If eye region **14** has one eye, region axis **17** may substantially coincide with eye axis **15**. If eye region **14** has two eyes, region axis **17** may pass through a midpoint between the eyes.

(9) As an overview of an example system, ophthalmic system **10** includes an eye camera **12**, an ophthalmic device **22**, a display **24**, a computer **26** (which includes logic **27** and memory **28**), and an illuminator **30**, coupled as shown. Eye camera **12** includes a camera system **20** and computer **26**, coupled as shown. As an overview of an example of operation, eye camera **12** provides images of an eye region **14**. Camera system **20** has cameras that yield image portions of eye region **14**. Each camera is located at a known position (e.g., a known location and/or orientation relative to each other and/or to eye region **14**) and records at least a portion of eye region **14** to yield an image portion. Computer **26** receives the image portions from the camera system **20**. Then, computer **26** identifies target pixels (e.g., reflection pixels) of one image portion at a location of the eye region, and determines image information of correction pixels of another image portion that image the same location of the eye region. Computer **26** corrects the target pixels using the identified image information. In some embodiments, if there are locations where no image provides usable (e.g., non-reflection) image data, data from surrounding pixels may be interpolated to provide information for these locations.

(10) Any suitable target pixels may be corrected. For example, a target pixel may be a reflection pixel that images a reflection of light from the eye and is corrected to reduce the image of the reflection. As another example, a target pixel may be an obstructed pixel that images a location of the eye that has been blocked by, e.g., an instrument or a body part such as an eyelash.

(11) Turning to the components, camera system **20** has a field of view (FOV) that covers eye region **14**. The FOV has a known relationship to the coordinate system of camera system **20**, which in certain embodiments has a known relationship to the coordinate system that ophthalmic device **22** uses to treat and/or diagnose an eye. In these embodiments, an eye tracker may track the position and movement of an eye by tracking the position and movement of the eye relative to the FOV. The eye tracking information may be used by ophthalmic device **22** to treat and/or diagnose the eye.

(12) In the embodiments, camera system **20** includes cameras. For ease of explanation, the “position” of a camera relative to eye region **14** may describe the distance between the camera and eye region **14** and the direction of the camera axis relative to region axis **17**. A camera detects light from an object and generates a signal in response to the light. The signal carries image data that can be used to generate the image of the eye. The image data are provided to computer **26** for eye tracking (and optionally other analysis) and may also be provided to display **24** to present the images of the eye. Examples of cameras include a charged-coupled device (CCD), video,

complementary metal-oxide semiconductor (CMOS) sensor (e.g., active-pixel sensor (APS)), line sensor, and optical coherence tomography (OCT) camera.

(13) A camera detects light of any suitable spectral range, e.g., a range of infrared (IR), ultraviolet (UV), and/or visible (VIS) wavelength light, where a range can include a portion or all of the wavelength. For example, a camera may detect visible light, infrared light, or other visible and infrared light from eye region **14** to yield an image portion. Certain cameras may capture features of the eye (e.g., pupil, iris, blood vessels, limbus, sclera, eyelashes, and/or eyelid) better than others. For example, an infrared camera generally provides more stable pupil tracking and better contrast for iris structures. Accordingly, an IR camera may be used to monitor lateral movement by tracking the pupil and/or cyclotorsion by tracking iris structures. As another example, a visible range camera yields better images of blood vessels, so a visible range camera may be used to monitor translation and/or rotational movement by tracking blood vessels.

(14) A camera may record images at any suitable frequency or resolution. A higher speed camera may record images at greater than, e.g., 400 to 1500 frames per second, such as greater than 500, 750, or 1000 frames per second. A higher resolution camera may yield images with greater than, e.g., 4 to 24 megapixels, such as greater than 5, 10, 15, or 20 megapixels. In general, higher resolution images and higher speed image acquisition may provide more accurate tracking, but both features may require more computing time, so there may be a trade-off between resolution and speed. Accordingly, the speed and/or resolution of a camera may be selected for particular purposes. In certain embodiments, a higher speed camera may track eye features that move faster and/or can be identified with lower resolution, and a higher resolution camera may be used to track eye features that require higher resolution for identification and/or move more slowly. For example, a lower resolution, higher speed camera may track the pupil (which does not require high resolution) to detect xy-movement. As another example, a higher resolution, lower speed camera may track blood vessels/iris structures to detect rotations, z-movement.

(15) Ophthalmic device **22** may be a system that is used to diagnose and/or treat an eye. Examples include a refractive surgical system, a cataract system, a topographer, an OCT measuring device, and a wavefront measuring device. Display **24** provides images to the user of system **10**. Examples of display **24** include a computer monitor, a 3D display, a projector/beamer, a TV monitor, binocular displays, glasses with monitors, a virtual reality display, an augmented reality display, and a mixed reality display.

(16) Illuminator **30** directs light towards eye region **14** to illuminate the eye for imaging. An illuminator **30** may comprise one or more light sources, such as one or more of any of the following, a lamp, an LED (which may be white or monochrome, e.g., green, red, IR, or UV), a laser diode (with the same example colors as the LED), and/or a projected light pattern (e.g., dots, lines, or crosses). In certain embodiments, room illumination or sunlight may provide illumination.

(17) Computer **26** controls components of system **10** (e.g., camera system **20**, an ophthalmic device **22**, a display **24**, and/or light projector **30**) to image an eye. In general, computer **16** receives the image portions from camera system **20** and corrects pixels of the image portions to yield an image of eye region **14**. As an overview, computer **26** identifies target pixels, e.g., reflection pixels that image reflection of light in an image. Computer **26** determines image information of correction pixels from, e.g., another image, and corrects the target pixels using the image information from the correction pixels.

(18) In embodiments that correct reflection pixels, the reflection pixels may be identified in any suitable manner. For example, computer **26** may use image processing to detect light saturated pixels at the reflection pixels. A light saturated pixel may be, e.g., a pixel with a greater than 90 percent of maximum level reading. As another example, computer **26** may calculate the location of the reflection pixels according to the position of illuminator **30** and the position of the camera that provided the image. The position of illuminator **30** provides the direction of the light rays incident on and reflected from the eye. The position of the camera provides the location of the eye where

the camera received the reflected light. As another example, computer **26** may receive eye tracking information describing movement of the eye, and then calculate the location of the reflection pixels according to the previous location of the pixels and the movement of the eye. Since the reflection moves with the eye, the location of the reflection pixels can be determined from the previous location and the movement.

(19) In the embodiments, computer **26** determines image information using correction pixels of a second image portion, where the correction pixels image the same location of the eye where the reflections appear in the first image. Image information may be determined in any suitable manner. For example, the image information may be determined from a second image portion provided by a different camera in a different position from the camera that provided the first image portion, e.g., stereoscopically arranged cameras. Since the cameras are in different positions, a reflection may appear at a location of the eye in one image, but the reflection may appear at different location of the eye in another image.

(20) As another example, the image information may be determined from a second image portion provided by the same camera that provided the first image portion, where the camera generated the first and second image portions at different times. As another example, computer **26** may determine the image information from a second image portion that comprises default eye image information. Default eye image information may comprise, e.g., image data of a typical eye at the location. For example, if the correction pixels are for correcting pixels at the pupil, the default eye image information may comprise dark pixels.

(21) In the embodiments, computer **26** uses the identified image information to reduce reflection in any suitable manner. For example, computer **26** may replace the reflection pixels with correction pixels. As another example, computer **26** may apply an averaging function to the reflection pixels and correction pixels. For example, the averaging function may weight the pixels equivalently or may give greater weight to pixels from, e.g., an image with higher quality. As another example, computer **26** may generate a correction overlay using the correction pixels, and place the correction overlay over the reflection pixels.

(22) FIG. 2 illustrates an example of the field of view (FOV) **40** of camera system **20** of FIG. 1, according to certain embodiments. The cameras of camera system **20** may have any suitable arrangement. For example, Camera A and Camera A may be arranged mirror symmetrically about system axis **14**, i.e., spatially separated with equal viewing angles on opposite sides of system axis **14**. The images may be stereoscopically reconstructed to track the location and orientation of an eye in two or three dimensions. The greater the angle and/or distance between the cameras, the better the accuracy in the z-direction. This may facilitate positioning the head of the patient.

(23) A camera of camera system **20** has a field of view (FOV) that detects light from eye region **14** to yield an image portion **45** of some or all of eye region **14**. Different cameras can have different FOVs that detect light from different portions of eye region at different directions, and different FOVs may overlap. In certain embodiments, the combined FOVs from the cameras yield a system FOV **40**. In general, more cameras at different positions (locations and orientations) may improve the detection of eye features and the accuracy of the tracking.

(24) In the example, camera system **20** has a system FOV **40**, a system axis **42**, and a system coordinate system **44** (x' , y' , z'). System axis **42** may have any suitable position, e.g., axis **42** may be substantially orthogonal to system FOV **40** and may pass through the center of system FOV **40**. System axis **42** and system coordinate system **44** (x' , y' , z') may be related in any suitable manner. In the example, system axis **42** defines the z' -axis of system coordinate system **44**. In the example, system FOV **40** is generally planar and images the numbers **1** through **9**. Camera system **20** includes Camera A with FOV A and Camera B with FOV B, where FOV A and FOV B extend partially or fully over system FOV **40**. Camera A provides a first image portion taken with FOV A, and Camera B provides a second image portion taken with FOV B.

(25) In certain embodiments, computer **26** aligns and combines image portions **45** to yield

combined image **46**. Image portions **45** may be aligned in any suitable manner. For example, each camera has a known position, such as a location (e.g., distance away from system FOV **40** and/or eye region **14**), orientation (e.g., camera optical axis relative to system axis **42** and/or eye axis **15**, or viewing angle), dimensions, and imaging properties. From this information, computer **26** can determine the positions of image portions **45** to align them within combined image **46**. As another example, the cameras each generate an image of a calibration figure (e.g., a checkerboard), and the positions of the cameras are determined from the images. As yet another example, a user calibrates image portions **45** by manually aligning portions **45** when viewed through the cameras. Computer **26** records the positions of the aligned portions.

(26) FIGS. **3A** and **3B** illustrate examples camera system **20** of FIG. **1** tracking eye regions **14**, according to certain embodiments. In FIG. **3A**, eye region **14** includes one eye. In the example, eye axis **15** of the eye may at first be substantially aligned with system axis **42** of camera system **20**. As the eye moves relative to camera system **20**, eye axis **15** moves relative to system axis **42**.

(27) In FIG. **3B**, eye region **14** includes both eyes. System axis **42** of camera system **20** may be aligned in any suitable location, e.g., substantially aligned with the midpoint between the eyes. Camera system **20** includes cameras that image one or both eyes to yield a combined image that images both eyes simultaneously, so camera system **20** can track both eyes simultaneously and independently of one another. In certain embodiments, camera system **20** includes a pair of stereoscopic cameras that can each image both eyes to provide three-dimensional image information, including z-depth information for both eyes.

(28) FIGS. **4** and **5** illustrate an example of a method for correcting pixels, such as reflection pixels, in an eye image, according to certain embodiments. FIG. **4** shows image portions A and B that are used to yield corrected image **46**, and FIG. **5** shows a flowchart of the method. The method starts at step **110**, where illuminator **30** directs light towards eye region **14**. Camera system **20** generates image portions A and B of eye region **14** at step **112**. Computer **26** receives image portions A and B from camera system **20** at step **114**.

(29) Computer **26** identifies reflection pixels of image portion A at step **116**. The reflection pixels image the reflection of light at a particular location of eye region **14**. Reflection pixels may be identified in any suitable manner. For example, computer **26** may detect light saturated pixels as reflection pixels. As another example, computer **26** may calculate the location of the reflection pixels according to the position of illuminator **30** and position of the camera that provided image portion A. As another example, computer **26** may receive eye tracking information describing movement of the eye, and then calculate the location of the reflection pixels according to the previous position of the pixels and the movement of the eye.

(30) Computer **26** determines image information of correction pixels of image portion B that image the same location of the eye region at step **120**. Image information may be determined in any suitable image portion B. For example, image portion B may be provided by a camera distinct from the camera that provided image portion A. As another example, image portion B may be provided by the same camera, but image portion B may have been taken before or after image portion A. As another example, image portion B may comprise default eye image information.

(31) Computer **26** corrects the reflection pixels using the identified image information to reduce reflection at step **122**. The reflection pixels may be corrected in any suitable manner. For example, computer **26** may replace the reflection pixels with correction pixels. As another example, computer **26** may apply an averaging function to the reflection pixels and correction pixels. As another example, computer **26** may generate a correction overlay using the correction pixels, and place the correction overlay onto the reflection pixels.

(32) A component (such as the control computer) of the systems and apparatuses disclosed herein may include an interface, logic, and/or memory, any of which may include computer hardware and/or software. An interface can receive input to the component and/or send output from the component, and is typically used to exchange information between, e.g., software, hardware,

peripheral devices, users, and combinations of these. A user interface is a type of interface that a user can utilize to communicate with (e.g., send input to and/or receive output from) a computer. Examples of user interfaces include a display, Graphical User Interface (GUI), touchscreen, keyboard, mouse, gesture sensor, microphone, and speakers.

(33) Logic can perform operations of the component. Logic may include one or more electronic devices that process data, e.g., execute instructions to generate output from input. Examples of such an electronic device include a computer, processor, microprocessor (e.g., a Central Processing Unit (CPU)), and computer chip. Logic may include computer software that encodes instructions capable of being executed by an electronic device to perform operations. Examples of computer software include a computer program, application, and operating system.

(34) A memory can store information and may comprise tangible, computer-readable, and/or computer-executable storage medium. Examples of memory include computer memory (e.g., Random Access Memory (RAM) or Read Only Memory (ROM)), mass storage media (e.g., a hard disk), removable storage media (e.g., a Compact Disk (CD) or Digital Video or Versatile Disk (DVD)), database, network storage (e.g., a server), and/or other computer-readable media. Particular embodiments may be directed to memory encoded with computer software.

(35) Although this disclosure has been described in terms of certain embodiments, modifications (such as changes, substitutions, additions, omissions, and/or other modifications) of the embodiments will be apparent to those skilled in the art. Accordingly, modifications may be made to the embodiments without departing from the scope of the invention. For example, modifications may be made to the systems and apparatuses disclosed herein. The components of the systems and apparatuses may be integrated or separated, or the operations of the systems and apparatuses may be performed by more, fewer, or other components, as apparent to those skilled in the art. As another example, modifications may be made to the methods disclosed herein. The methods may include more, fewer, or other steps, and the steps may be performed in any suitable order, as apparent to those skilled in the art.

(36) To aid the Patent Office and readers in interpreting the claims, Applicants note that they do not intend any of the claims or claim elements to invoke 35 U.S.C. § 112(f), unless the words “means for” or “step for” are explicitly used in the particular claim. Use of any other term (e.g., “mechanism,” “module,” “device,” “unit,” “component,” “element,” “member,” “apparatus,” “machine,” “system,” “processor,” or “controller”) within a claim is understood by the applicants to refer to structures known to those skilled in the relevant art and is not intended to invoke 35 U.S.C. § 112(f).

Claims

1. An ophthalmic system that images an eye region comprising at least one eye, comprising: a camera system comprising a plurality of cameras configured to yield a plurality of image portions of the eye region, each camera located at a position relative to the eye region and configured to yield an image portion of the plurality of image portions; an illuminator located at a position relative to the eye region and configured to direct light towards the eye region; and a computer configured to: receive the plurality of image portions from the camera system, the plurality of image portions comprising a first image portion and a second image portion, the first image portion provided by a first camera, the second image portion provided by a second camera; identify one or more reflection pixels of the first image portion, the reflection pixels imaging a reflection of light from the illuminator reflected by a location of the eye region; determine image information of one or more correction pixels of the second image portion, the correction pixels imaging the same location of the eye region; and correct the reflection pixels using the identified image information to reduce the reflection by: generating a correction overlay using the one or more correction pixels; and placing the correction overlay over the one or more reflection pixels.

2. The ophthalmic system of claim 1, the plurality of cameras comprising a set of stereoscopic cameras arranged symmetrically about a system axis of the camera system.
3. The ophthalmic system of claim 1, the computer configured to identify one or more reflection pixels of the first image portion by: detecting one or more light saturated pixels as the one or more reflection pixels.
4. The ophthalmic system of claim 1, the computer configured to identify one or more reflection pixels of the first image portion by: identifying the one or more reflection pixels according to the position of the illuminator relative to the eye region and the position of the first camera relative to the eye region.
5. The ophthalmic system of claim 1, the computer configured to identify one or more reflection pixels of the first image portion by: receiving eye tracking information describing movement of the eye; and identifying the one or more reflection pixels according to a previous location of the one or more reflection pixels and the movement of the eye.
6. The ophthalmic system of claim 1, the first camera distinct from the second camera, the first camera at a first position relative to the eye region, the second camera at a second position relative to the eye region distinct from the first position.
7. The ophthalmic system of claim 1, the first camera the same as the second camera, the second image portion provided prior to the first image portion.
8. The ophthalmic system of claim 1, the second image portion comprising default eye image information.
9. The ophthalmic system of claim 1, the computer configured to correct the reflection pixels using the identified image information by: replacing the one or more reflection pixels with the one or more correction pixels.
10. The ophthalmic system of claim 1, the computer configured to correct the reflection pixels using the identified image information by: applying an averaging function to the one or more reflection pixels and the one or more correction pixels.
11. An ophthalmic system that images an eye region comprising at least one eye, comprising: a camera system comprising a plurality of cameras configured to yield a plurality of image portions of the eye region, each camera located at a position relative to the eye region and configured to yield an image portion of the plurality of image portions; and a computer configured to: receive the plurality of image portions from the camera system, the plurality of image portions comprising a first image portion and a second image portion, the first image portion provided by a first camera, the second image portion provided by a second camera; identify one or more target pixels of the first image portion, the target pixels imaging a location of the eye region; determine image information of one or more correction pixels of the second image portion, the correction pixels imaging the same location of the eye region; and correct the target pixels using the identified image information by: generating a correction overlay using the one or more correction pixels; and placing the correction overlay over the one or more target pixels.
12. The ophthalmic system of claim 11, the plurality of cameras comprising a set of stereoscopic cameras arranged symmetrically about a system axis of the camera system.
13. The ophthalmic system of claim 11, the computer configured to identify one or more target pixels of the first image portion by: detecting one or more light saturated pixels as the one or more target pixels.
14. The ophthalmic system of claim 11, the computer configured to identify one or more target pixels of the first image portion by: receiving eye tracking information describing movement of the eye; and identifying the one or more target pixels according to a previous location of the one or more target pixels and the movement of the eye.
15. The ophthalmic system of claim 11, the first camera distinct from the second camera, the first camera at a first position relative to the eye region, the second camera at a second position relative to the eye region distinct from the first position.

16. The ophthalmic system of claim 11, the first camera the same as the second camera, the second image portion provided prior to the first image portion.
 17. The ophthalmic system of claim 11, the second image portion comprising default eye image information.
 18. The ophthalmic system of claim 11, the computer configured to correct the target pixels using the identified image information by: replacing the one or more target pixels with the one or more correction pixels.
 19. The ophthalmic system of claim 11, the computer configured to correct the target pixels using the identified image information by: applying an averaging function to the one or more target pixels and the one or more correction pixels.
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